DUBLIN CITY UNIVERSITY

SEMESTER ONE EXAMINATIONS 2024/25

MODULE:	MTH1034: Linear algebra MTH1034
QUALIFICATIONS:	AP BSc Applied Physics CASE BSc Computer Apps (Sft.Eng.) COMSCI BSc Computer Science PBM BSc Physics Biomedical Sci PHA BSc Physics with Astronomy PAN BSc Physics with Data Analytic PEM BSc Physical Education and Maths AI Chemistry with AI
YEAR OF STUDY:	1
EXAMINERS:	Dr. Abraham Harte (Internal: Ext. 5669) Dr. Niall Madden (External)
TIME ALLOWED:	2 hours
INSTRUCTIONS:	Attempt all questions. Note that questions do not carry equal marks.
REQUIREMENTS:	Two self-written A4 pages may be used as notes. Calculators may be used as well.

Advice: Please organise your work, clearly marking your final answers. Marks will not be awarded for irrelevant or contradictory statements which make no significant progress towards a solution.

Most parts of the multi-part questions can be answered independently, at least if the preceding parts have been at least been read. If you get stuck on something, move on to the next part and return later if you have time.

QUESTION 1 [TOTAL MARKS: 14]

Suppose that there are three aircraft, the locations of which are given by

$$\boldsymbol{x}_1 = \begin{pmatrix} 10 \\ 20 \\ 2 \end{pmatrix}, \qquad \boldsymbol{x}_2 = \begin{pmatrix} 5 \\ 21 \\ 6 \end{pmatrix}, \qquad \boldsymbol{x}_3 = \begin{pmatrix} 20 \\ 30 \\ 3 \end{pmatrix}$$

at some moment in time.

(a) What is the angle between the second and third aircraft as seen by the first?

[6 marks]

(b) If aircraft 1 has the velocity $\boldsymbol{v} = (4, 2, 0)$, find the *lateral* separation of aircraft 2 from aircraft 1. This is the distance between the two along a direction which is orthogonal to $\boldsymbol{v}$.

[8 marks]

QUESTION 2 [TOTAL MARKS: 28]

Let A be a matrix which rotates vectors in $\mathbb{R}^2$ counterclockwise by $45°$. Let B be a matrix which projects vectors in $\mathbb{R}^2$ onto the line parallel to $(1, 1)$.

(a) If $\boldsymbol{x} = (1, 1)/\sqrt{2}$, sketch $\boldsymbol{x}$, $A\boldsymbol{x}$, $B\boldsymbol{x}$, $AB\boldsymbol{x}$, and $BA\boldsymbol{x}$.

[10 marks]

(b) Find all positive integers p such that $A^p = I$.
[Hint: It may help to think about this geometrically.]

[8 marks]

(c) Find A and B.

[10 marks]

QUESTION 3 [TOTAL MARKS: 18]

A sensor has two input lines and two output lines. Its output $\boldsymbol{f} : \mathbb{R}^2 \to \mathbb{R}^2$ is known to be a linear function of its inputs.

(a) An experiment shows that

$$\boldsymbol{f}(\boldsymbol{i}_1) = \boldsymbol{o}_1, \qquad \boldsymbol{f}(\boldsymbol{i}_2) = \boldsymbol{o}_2,$$

where $\boldsymbol{i}_1$ and $\boldsymbol{i}_2$ are two particular inputs and $\boldsymbol{o}_1$ and $\boldsymbol{o}_2$ are the measured outputs. State a general property which $\boldsymbol{i}_1$ and $\boldsymbol{i}_2$ can have which would guarantee that these two measurements completely characterize the function $\boldsymbol{f}$ (for arbitrary inputs). Justify your answer.

[8 marks]

(b) Suppose that for all $\boldsymbol{x} \in \mathbb{R}^2$,

$$\boldsymbol{f}(\boldsymbol{x}) = \begin{pmatrix} 3 & 1 \\ 6 & 2 \end{pmatrix} \boldsymbol{x}.$$

If the output is measured to be $(2, 4)$, what exactly can be said about the input?

[10 marks]

QUESTION 4 [TOTAL MARKS: 20]

A ball is dropped and its height is measured at the n times t_i, where $i = 1, \ldots, n$. This yields the data (t_i, h_i). The ball is expected to fall with the constant acceleration a, meaning that its height is expected to satisfy

$$h(t) = x_0 + v_0 t - \frac{1}{2} a t^2,$$

where x_0 and v_0 are also constants. Least-squares regression can be used to fit this model to the data using

$$\begin{pmatrix} x_0 \\ v_0 \\ -\frac{1}{2}a \end{pmatrix} = (M^T M)^{-1} M^T \begin{pmatrix} h_1 \\ \vdots \\ h_n \end{pmatrix},$$

where M is an $n \times 3$ matrix.

(a) How should M be constructed from the data?

[8 marks]

(b) The data (t_i, h_i) is very unlikely to exactly match the model $h(t)$; there will almost inevitably be some scatter around the "best-fit" curve. What exactly is "best" about the above calculation of x_0, v_0, and a?

[6 marks]

(c) Suppose that someone tells you that the right-hand side of the above regression formula can be simplified using

$$\left(M^T M\right)^{-1} M^T = \left(M^{-1}(M^T)^{-1}\right) M^T = M^{-1}\left((M^T)^{-1} M^T\right) = M^{-1}.$$

What is wrong with this argument? Assume that $n \neq 3$.

[6 marks]

QUESTION 5 [TOTAL MARKS: 20]

A 2×2 matrix B is known to satisfy

$$B \begin{pmatrix} 1 \\ 3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}, \qquad B \begin{pmatrix} -1 \\ 1 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} -1 \\ 1 \end{pmatrix}.$$

(a) Compute $\det B$.

[6 marks]

(b) Find B^n for all positive integers n.

[14 marks]